Bluestock Fintech

Mutual Fund Analytics

Final Project Report

Data Engineering • Exploratory Analysis • Advanced Analytics • Power BI Dashboard

Project period: Internship project

Report version: 1.0

Prepared as the final analytical report for the Mutual Fund Analytics project.

1. Executive Summary

This project develops an end-to-end Mutual Fund Analytics solution for Bluestock Fintech. The workflow covers raw-data ingestion, cleaning and validation, exploratory data analysis, fund-performance measurement, investor and SIP analytics, advanced risk metrics, portfolio concentration analysis, recommendation logic, and dashboard reporting.

The analytical layer was implemented primarily with Python and pandas, with SQL/SQLite used for structured querying and data storage. Power BI was used to convert the processed datasets into interactive management dashboards. The final project repository separates raw data, processed outputs, notebooks, reports, SQL assets, and reusable Python scripts.

Major outcomes

- Created a reproducible data-cleaning and processing workflow for mutual-fund datasets.
- Produced processed analytical datasets covering NAV history, fund performance, investor transactions, SIP continuity, risk metrics, and portfolio concentration.
- Built advanced analytics for historical VaR/CVaR, rolling Sharpe ratio, investor cohorts, SIP continuation, recommendation logic, and sector HHI.
- Developed a four-page Power BI dashboard covering industry overview, investor analytics, fund performance, and SIP/market trends.
- Organized the final project outputs into a submission-ready repository structure.

Management-level takeaway: The dashboard brings fund-house scale, investor behavior, performance, SIP activity, and risk indicators into one reporting workflow, enabling faster comparison and more consistent decision support.

2. Project Objectives

The project was designed around five practical objectives:

- Measure the scale and growth of the mutual-fund industry using AUM, folio and fund-house information.
- Compare fund performance using return, volatility and risk-adjusted metrics.
- Understand investor behavior through transaction, demographic, cohort and SIP analysis.
- Identify risk using historical VaR/CVaR, rolling Sharpe and portfolio concentration measures.
- Present results through a business-oriented Power BI dashboard and supporting analytical reports.

Target deliverables

- Cleaned and validated CSV datasets.
- Python notebooks and reusable scripts.
- SQL schema and analytical queries.
- Advanced Analytics notebook and supporting outputs.
- Power BI dashboard screenshots.
- Final report and presentation-ready project material.

3. Data Sources and Repository Architecture

The repository uses a clear raw-to-processed structure. Raw files are retained as source datasets while processed files contain cleaned or derived analytical outputs. This separation reduces the risk of overwriting source data and makes the workflow easier to audit.

Raw data included in the project

- Fund master data.
- NAV history.
- AUM by fund house.
- Monthly SIP inflows.
- Category inflows.
- Industry folio count.
- Scheme performance.
- Investor transactions.
- Portfolio holdings.
- Benchmark indices.
- Live NAV datasets for selected schemes.

Processed analytical outputs

- daily_returns.csv
- fund_scorecard.csv
- alpha_beta.csv
- CAGR_Comparison.csv
- sharpe_ratio.csv / sortino_ratio.csv
- var_cvar_report.csv
- cohort_analysis.csv
- sip_continuity.csv
- sector_hhi.csv
- tracking_error.csv

4. Data Cleaning and ETL

The ETL stage transforms source CSV files into analysis-ready datasets. Python/pandas is used for loading, cleaning, type conversion, date handling, validation and derived columns. The project also contains dedicated scripts such as clean_investor_transactions.py, clean_nav_history.py, clean_scheme_performance.py, data_ingestion.py and validate_data.py.

Typical ETL sequence

- Load source CSV files from Data/Raw.
- Inspect column names, data types, missing values and duplicate records.
- Standardize dates, numerical fields and categorical values.
- Create derived analytical fields such as year, month, returns and investor cohort indicators.
- Validate the cleaned datasets before downstream analysis.
- Write outputs to Data/Processed.
- Retain reusable code so the workflow can be rerun when source data changes.

Quality-control principle

An important lesson during development was to verify actual column names before applying groupby, aggregation or filtering operations. This avoided assuming fields such as transaction_amount_lakh or scheme_name existed when the available dataset used amount_inr, amfi_code or other actual fields.

5. Exploratory Data Analysis

EDA was performed across industry scale, fund-house activity, fund performance, investor demographics, SIP behavior and portfolio composition. The analysis was intentionally split into focused notebooks rather than one large script, improving readability and traceability.

EDA areas covered

- AUM growth and fund-house comparison.
- Folio-count growth.
- NAV trends.
- Benchmark comparison.
- Category-level inflows.
- Investor age distribution.
- SIP amount by age group and state.
- Return/volatility relationships.
- Sector allocation and concentration.

Observed dashboard patterns

- SBI Mutual Fund appears as the largest fund house by AUM in the dashboard comparison.
- The industry AUM trend rises strongly through 2024 before declining in the displayed 2025 point.
- Monthly transaction activity reaches a high around the middle of the year and then moderates.
- SIP inflow shows an overall upward trend across the displayed 2022–2025 period.
- The category analysis highlights Liquidity as the largest displayed category by net inflow.

6. Fund Performance Analysis

Fund performance was evaluated using multiple horizons rather than relying on a single return number. The processed scheme-performance data contains 1-year, 3-year and 5-year returns together with risk and risk-adjusted fields such as standard deviation, Sharpe ratio, Sortino ratio, alpha, beta and maximum drawdown.

Performance metrics

• 1-year, 3-year and 5-year return percentages for horizon comparison.

• Standard deviation as a measure of return variability.

• Sharpe ratio for return relative to volatility.

• Sortino ratio with greater emphasis on downside variability.

• Alpha for excess performance relative to a benchmark model.

• Beta for sensitivity to benchmark movement.

• Maximum drawdown for downside-loss assessment.

The Power BI fund-performance page combines a NAV comparison, fund-house selector, risk/return scatter view and detailed scheme table. This allows users to move from a high-level comparison to individual scheme-level values.

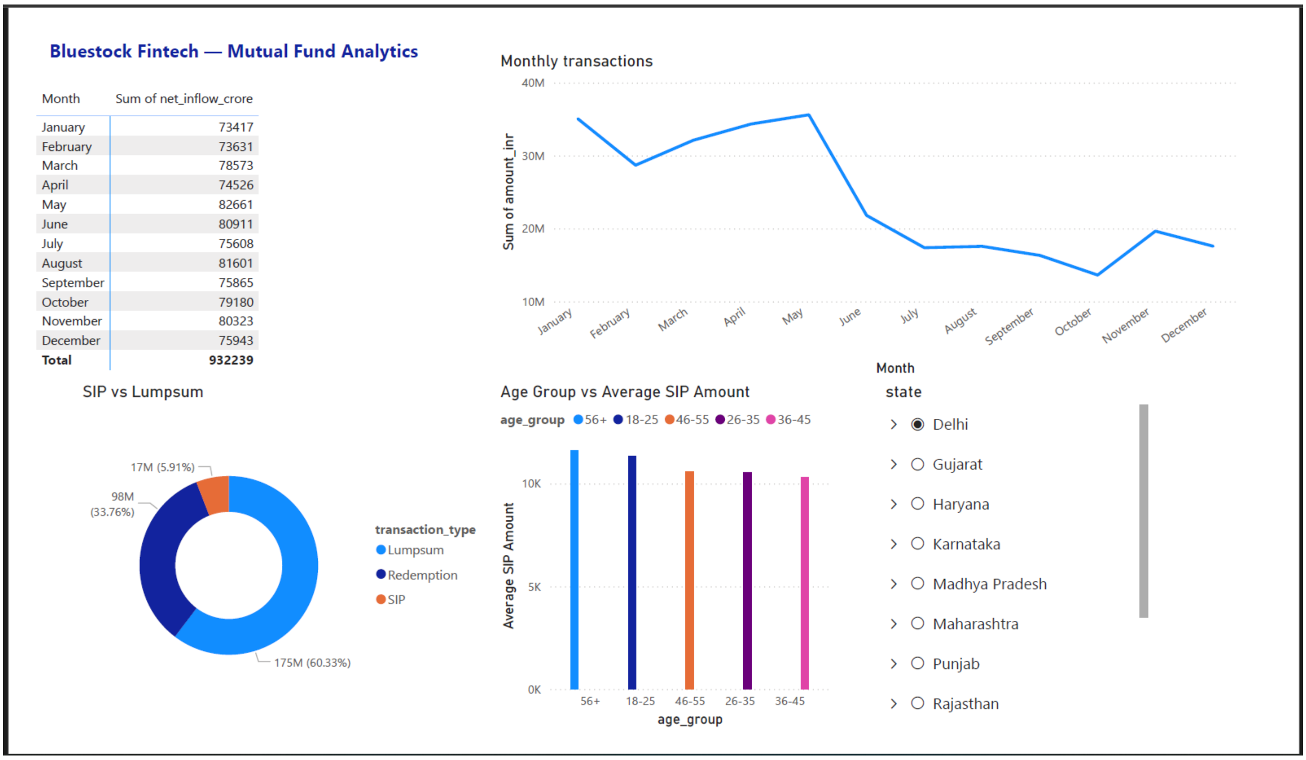

Figure 3. Power BI Dashboard — Fund Performance

7. Investor and Cohort Analysis

Investor analytics focuses on who is investing, how much they invest and how their activity changes over time. The transaction dataset contains investor identifiers, transaction dates, transaction types, amounts, location, age group, gender, income, payment mode and KYC status.

Cohort method

Investors are grouped by their first transaction year. For the 2024/2025 cohort analysis, the workflow filters investors whose first transaction falls within those years and then aggregates transaction activity at investor level. The final cohort output is stored in Data/Processed/cohort_analysis.csv.

Investor insights supported by the dashboard

- Average SIP amount can be compared across age groups.
- State-level selections allow geographic comparison.
- Monthly transaction activity provides a time-based view of investor flows.
- Transaction type composition separates SIP, redemption and lump-sum activity.
- Cohort analysis provides a basis for comparing newer investors with earlier cohorts.

scheme_name	fund_house	category	plan	Sum of return_1yr_pct	Sum of return_3yr_pct	Sum of return_5yr_pct	Sum of sharpe_ratio
ABSL Frontline Equity Fund - Regular - Growth	Aditya Birla Sun Life MF	Large Cap	Regular	14.82	13.78	12.86	0.98
ABSL Liquid Fund - Regular - Growth	Aditya Birla Sun Life MF	Liquid	Regular	6.18	5.14	7.95	5.14
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	Small Cap	Regular	24.93	22.38	23.80	0.90
Axis Bluechip Fund - Direct - Growth	Axis Mutual Fund	Large Cap	Direct	13.95	12.14	13.66	0.87
Axis Bluechip Fund - Regular - Growth	Axis Mutual Fund	Large Cap	Regular	11.82	11.84	14.14	0.85
Axis Midcap Fund - Regular - Growth	Axis Mutual Fund	Mid Cap	Regular	14.88	15.18	18.94	0.80
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	Small Cap	Regular	21.97	20.98	22.62	0.84
DSP Midcap Fund - Regular - Growth	DSP Mutual Fund	Mid Cap	Regular	14.12	17.16	19.00	0.90
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	Small Cap	Regular	20.20	20.08	20.61	0.80
DSP Top 100 Equity Fund - Regular - Growth	DSP Mutual Fund	Large Cap	Regular	11.96	12.82	12.35	0.92
HDFC Mid-Cap Opportunities Fund - Direct - Growth	HDFC Mutual Fund	Mid Cap	Direct	19.98	15.29	15.85	0.80
Total				**575.04**	**563.56**	**580.67**	**54.47**

Figure 2. Power BI Dashboard — Investor Analytics

8. SIP Analysis

Systematic Investment Plan analysis examines inflow trends, investor participation and continuation behavior. The project calculates monthly SIP inflows and evaluates whether investors with repeated SIP transactions continue to invest or develop long gaps between transactions.

SIP continuation analysis

- Identify investors with at least six SIP transactions.
- Calculate gaps between consecutive SIP transactions.
- Flag investors with gaps greater than 35 days as at-risk.
- Store the resulting analysis in sip_continuity.csv.

The dashboard also compares SIP amounts across age groups and displays state-level filtering. This makes the analysis useful for identifying segments where SIP engagement may be stronger or weaker.

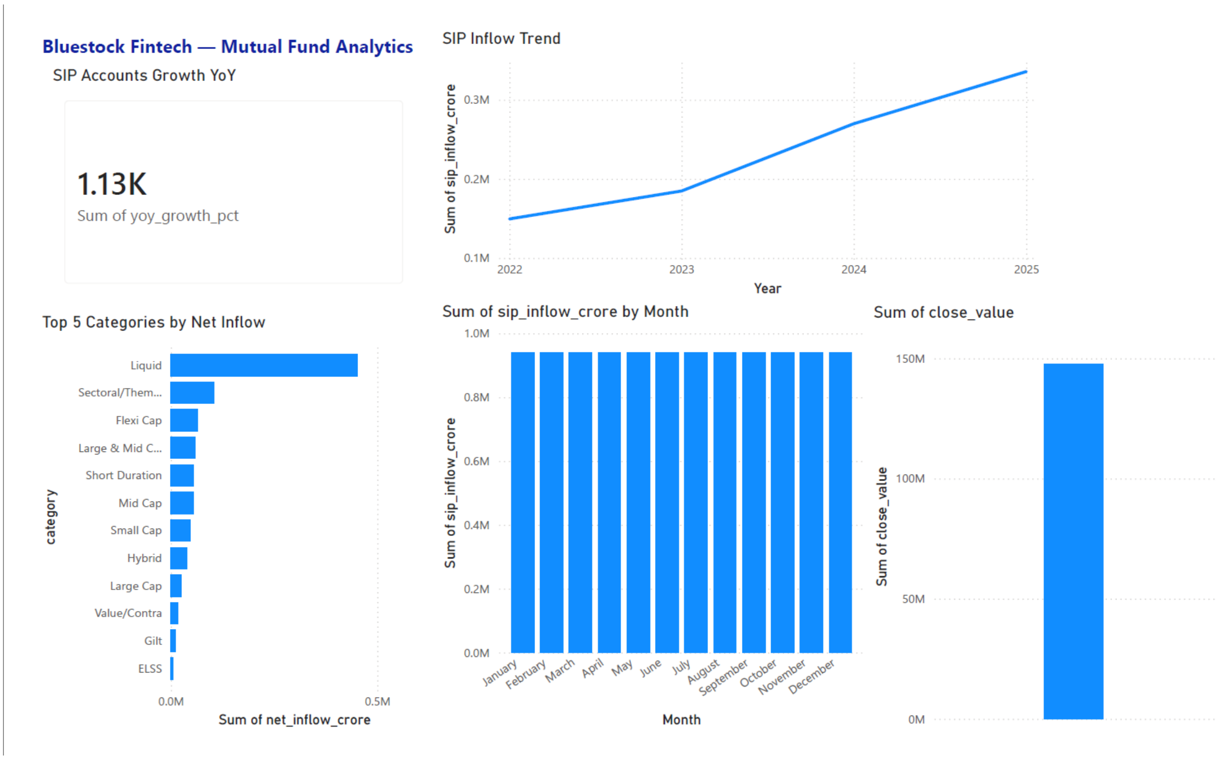

Figure 4. Power BI Dashboard — SIP & Market Trends

9. Advanced Risk Analytics

The advanced analytics stage adds risk metrics beyond simple return calculations. These measures help distinguish between funds that generate similar returns but carry different levels of downside or volatility.

Historical VaR (95%)

Historical Value at Risk uses the 5th percentile of the daily-return distribution. It provides a threshold below which only approximately the worst 5% of historical daily observations fall.

Conditional VaR

Conditional VaR, also called expected shortfall in many risk frameworks, takes the mean of returns below the VaR threshold. It therefore describes the average severity of losses in the tail rather than only the cutoff.

Rolling 90-day Sharpe

A rolling 90-day Sharpe calculation tracks risk-adjusted performance through time. The resulting chart helps identify periods when return generation became stronger or weaker relative to observed volatility.

Output

- var_cvar_report.csv
- rolling_sharpe_chart.png

10. Sector Concentration Analysis

Portfolio concentration is measured using the Herfindahl-Hirschman Index (HHI). For each equity fund, sector weights are squared and summed. A higher HHI indicates that portfolio weight is concentrated in fewer sectors, while a lower HHI indicates a more distributed allocation.

Formula

HHI = $\Sigma(w_{■}^{2})$, where $w_{■}$ represents the sector weight expressed consistently within the calculation.

The portfolio-holdings source contains AMFI code, stock symbol, stock name, sector, weight percentage, market value, current price and portfolio date. The resulting sector concentration file is stored as sector_hhi.csv, with a supporting visualization sector_hhi_chart.png.

Interpretation

- Higher HHI → greater sector concentration.
- Lower HHI → broader sector diversification.
- HHI should be interpreted alongside return and volatility rather than as a standalone recommendation measure.

11. Fund Recommendation Model

A simple rule-based recommendation model was developed to translate analytical metrics into an actionable screening output. The model accepts an investor risk preference such as Low, Moderate or High and returns top funds whose risk characteristics match the selected profile.

Recommendation logic

- Use the fund scorecard as the analytical base.
- Filter or rank funds according to risk grade.
- Consider return, volatility and risk-adjusted measures together.
- Return the top three matching funds for the selected risk profile.

The model is intentionally simple and transparent. It is a screening tool, not a personalized financial advisory system. Investors should consider objectives, time horizon, liquidity needs and professional advice before making actual investment decisions.

Deliverable: recommender.py

12. Power BI Dashboard

The Power BI layer converts the analytical outputs into a four-page management dashboard. The dashboard uses charts, KPI cards, tables, slicers and interactive selections to connect industry-level trends with fund-level and investor-level details.

Dashboard pages

- Page 1 — Industry Overview: AUM trend, total SIP inflows, total AUM, folio KPI and top fund houses.
- Page 2 — Investor Analytics: monthly transactions, transaction-type mix, SIP by age group and state slicer.
- Page 3 — Fund Performance: NAV comparison, fund-house selection, risk/return scatter and scheme table.
- Page 4 — SIP & Market Trends: SIP account growth, SIP inflow trend, top categories, monthly SIP inflow and market value.

13. Dashboard Screenshots

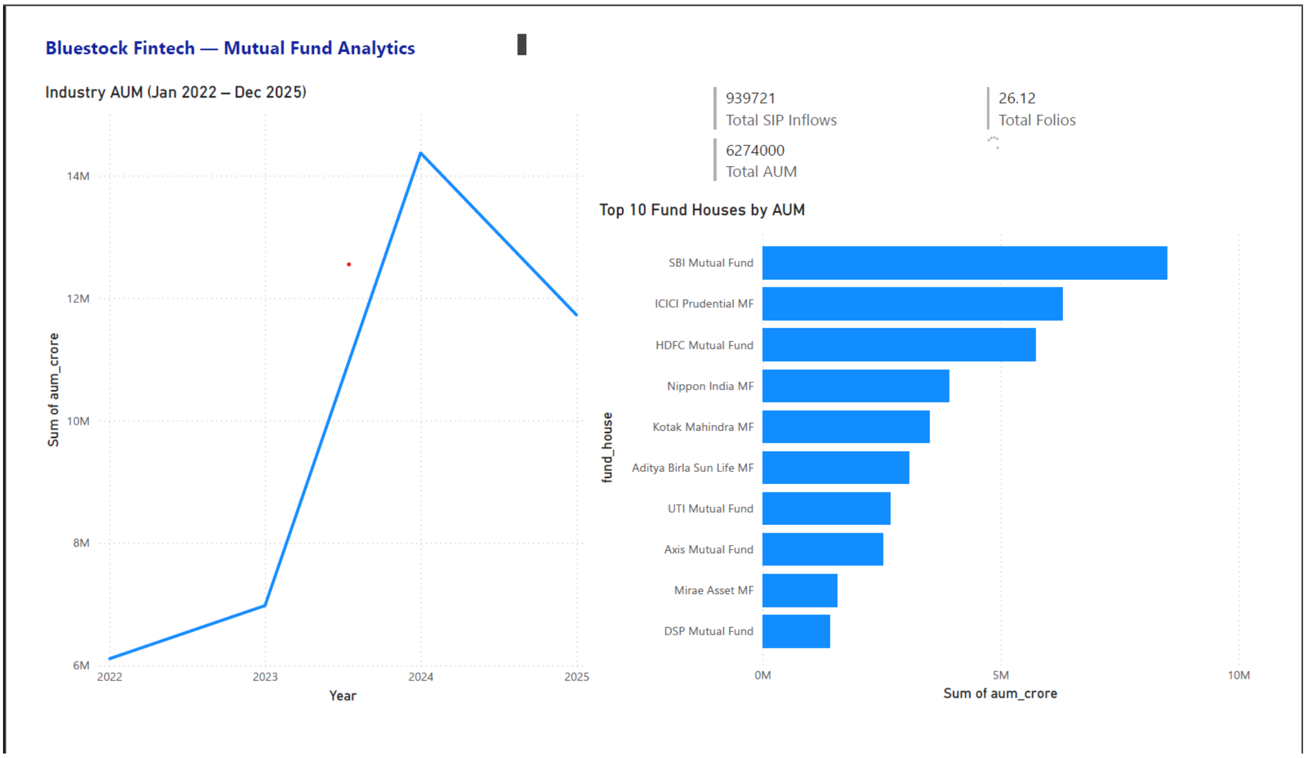

Figure 1. Power BI Dashboard — Industry Overview

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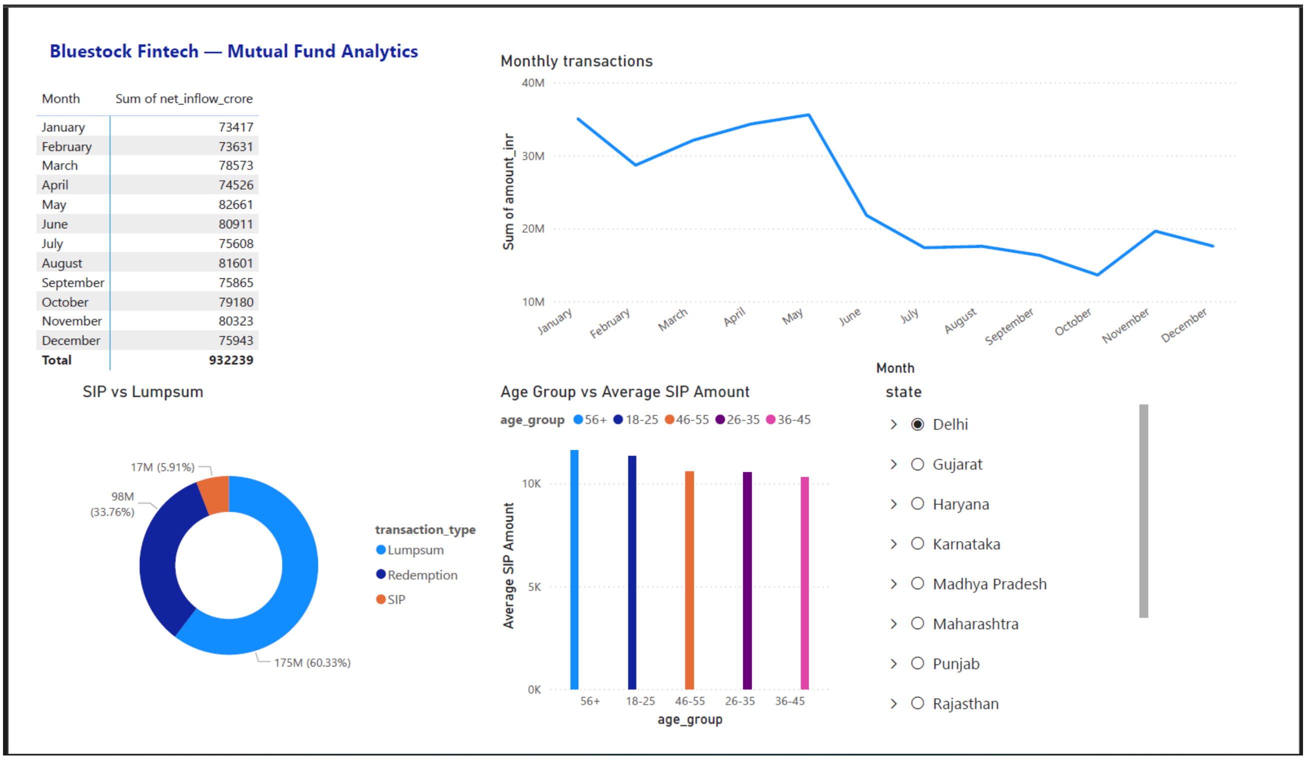

Figure 3. Power BI Dashboard — Fund Performance

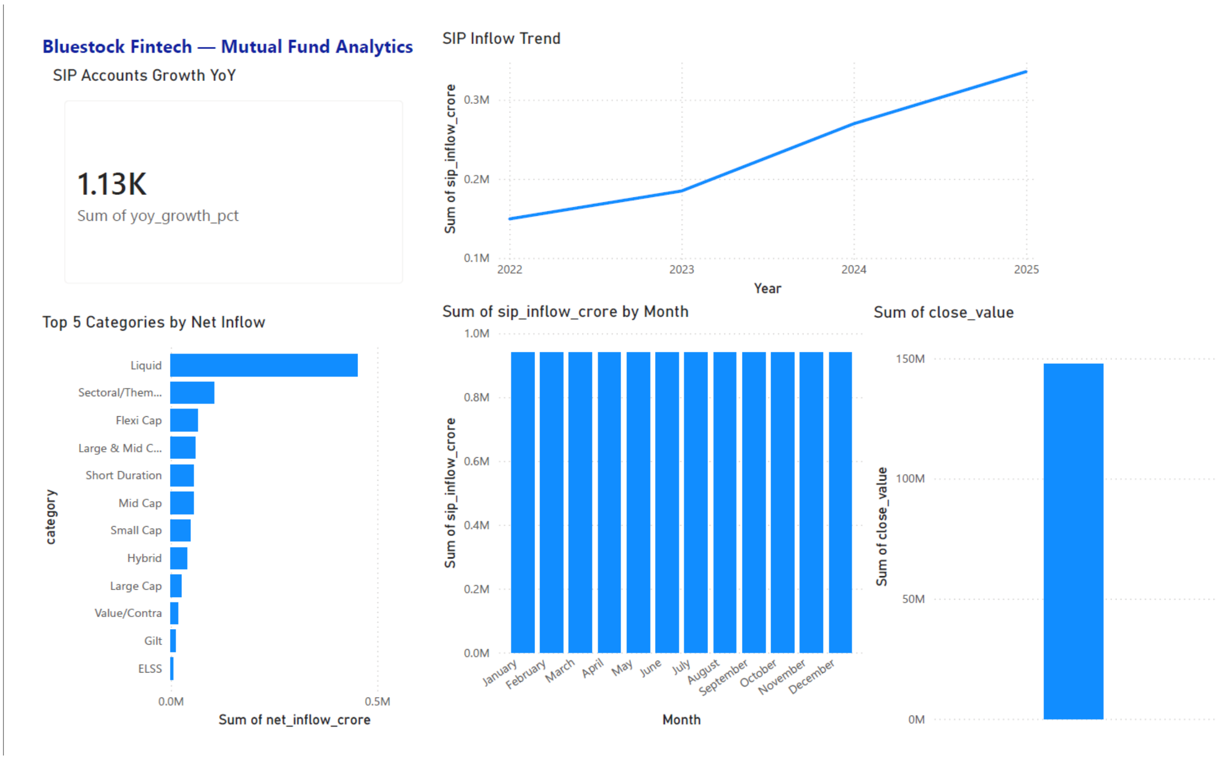

Figure 4. Power BI Dashboard — SIP & Market Trends

14. Key Findings

Industry and fund-house findings

- The dashboard indicates a strong rise in displayed industry AUM through 2024, followed by a lower displayed 2025 point.
- SBI Mutual Fund is the largest displayed fund house by AUM among the top-ten comparison.
- Fund-house scale varies substantially, making ranking useful for high-level market comparison.

Investor and SIP findings

- SIP activity can be segmented by age group and state, enabling targeted investor analysis.
- The dashboard's transaction mix shows SIP, redemption and lump-sum activity as distinct components.
- SIP inflow displays an upward trend over the displayed annual period.

Risk findings

- Risk-adjusted metrics provide a more informative comparison than raw returns alone.
- VaR/CVaR adds tail-risk information while rolling Sharpe adds a time-varying risk-adjusted perspective.
- HHI identifies concentration that may not be visible from return metrics.

15. Business Recommendations

- **Use a combined performance-risk score:** avoid ranking funds solely by historical return.
- **Monitor SIP continuation:** investors with repeated long gaps can be prioritized for engagement campaigns.
- **Segment investor communication:** use age and state dimensions to tailor educational and retention initiatives.
- **Track concentration:** funds with high HHI can be monitored for sector-specific concentration risk.
- **Automate refreshes:** keep raw and processed layers separate so new data can be incorporated without changing the dashboard structure.
- **Maintain metric definitions:** document units, time windows and formulas for every KPI to preserve consistency.

These recommendations are analytical and operational suggestions based on the project outputs. They should not be interpreted as individual investment advice.

16. Limitations

- Historical analysis does not guarantee future investment performance.
- Data coverage, frequency and quality depend on the supplied project datasets.
- The recommendation model is rule-based and does not model every investor-specific constraint.
- VaR and CVaR are historical estimates and depend strongly on the selected return window.
- HHI measures concentration but does not by itself measure the quality or risk of a sector.

17. Conclusion

The Mutual Fund Analytics project demonstrates a complete analytics workflow from raw financial datasets to business-facing decision support. The solution combines Python-based ETL and analysis, SQL/SQLite data management, advanced risk metrics and Power BI visualization. The resulting repository is structured so that data, code, notebooks, SQL and reporting assets can be maintained independently.

The strongest value of the project is the connection between descriptive and advanced analytics: industry scale and investor behavior are viewed alongside performance, risk and concentration. This creates a more complete analytical foundation for monitoring mutual-fund products and investor activity.

18. Tools, Technologies and Deliverables

Tools and technologies

- Python — pandas, NumPy, Matplotlib and supporting scripts.
- Jupyter Notebook — exploratory and advanced analytical workflows.
- SQL / SQLite — structured data storage and querying.
- Power BI — interactive dashboard development.
- VS Code — project organization and development.
- Git / GitHub — version control and project delivery.

Repository structure

- Data/Raw — original source datasets.
- Data/Processed — cleaned and derived analytical CSV files.
- Notebooks — analytical notebooks.
- Reports — dashboard screenshots, analytical charts and data dictionary.
- SQL — schema and queries.
- Root Python scripts — ingestion, cleaning, validation and recommendation utilities.

Key advanced outputs

- Advanced_Analytics.ipynb
- var_cvar_report.csv
- rolling_sharpe_chart.png
- cohort_analysis.csv
- sip_continuity.csv
- recommender.py
- sector_hhi.csv
- sector_hhi_chart.png

19. Appendix — Submission Checklist

Before final submission, verify the following:

- All processed CSV outputs are inside Data/Processed.
- All report images are inside Reports.
- All notebooks are inside Notebooks.
- SQL files are inside SQL.
- README.md is present at project root.
- requirements.txt is present.
- Python scripts execute without unresolved errors.
- Power BI dashboard pages load correctly.
- Final_Report.pdf is exported and opens correctly.
- Presentation deck is prepared.
- Git repository contains the final project state.